INTACT: Sound Worst-Case Topological Certificates for Segmentation Score Maps

Radoslav Lovecký
Arperon
info@arperon.com

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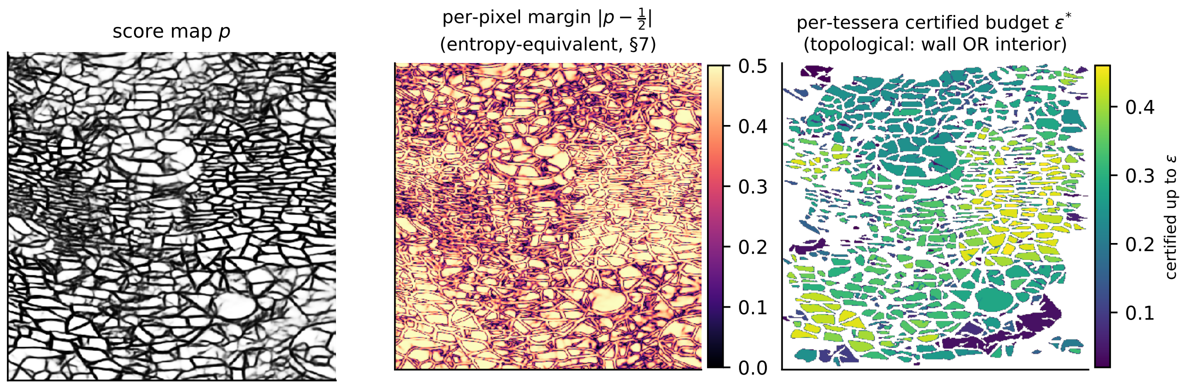

Figure 1: The certificate is not a confidence map. Left: a frozen score map p on a real mosaic crop. Middle: the per-pixel margin $|p - \frac{1}{2}|$ — a monotone transform of predictive entropy (§7), which calls almost every pixel confident. Right: what INTACT actually certifies — each enclosed tessera painted by ε^*, the largest ℓ_∞ budget at which it *provably* remains an unfillable hole. The budgets are *graded*: some tesserae survive to $\varepsilon{\approx}0.45$, others are lost by $\varepsilon{\approx}0.05$, because a hole dies when its *wall* erodes as readily as when its interior does — a topological fact that no per-pixel confidence map contains. Read off the exactly-nested sweep (§5.4), on a finer ε grid than Table 2; the crop is the one whose certified-cycle count is closest to the 30-crop cohort mean.

Abstract

Consumers of segmentation rarely ask about pixel accuracy; they ask topological questions — how many connected structures are there, does this line stay unbroken, how many enclosed regions does it bound. Topology-aware training methods (the clDice and Betti-matching lineage) improve such properties on average but guarantee nothing about any particular output. We present INTACT (INference-Time Adversarial Certificates for Topology): sound, worst-case topological certificates for thresholded segmentation score maps. An ℓ_∞ uncertainty band of radius ε around a frozen score map p determines — exactly — a lattice interval of reachable binary masks, between the certainly-foreground core CERT_FG $= \{p - \varepsilon > \frac{1}{2}\}$ and the complement of the certainly-background core CERT_BG $= \{p + \varepsilon \leq \frac{1}{2}\}$. Over this interval we prove a two-mask certification principle: a finite conjunction of monotone predicates holds on every reachable mask iff it holds at the interval's two extremes — and both extremes are individually necessary, a minimal check that no fixed budget of random sampling can replace. Instantiating the principle yields five certified quantities — certainly-foreground coverage, certified connected components, certified enclosed regions (tesserae), a strict certified-connectivity fraction, and certified 1-cycles — holes whose wall is certainly foreground and whose interior is certainly background, so no in-budget perturbation can fill or breach them — each stated with its worst-case adversary and its scope caveat. The certificate operates on perturbations of the score map; it is **not** an end-to-end input-space certificate (composition with input-space methods is an interface we state, not a result we claim).

The instantiation is not mechanical: our first cycle certificate, β_1 of the certified foreground, was genuinely unsound (H_1 is not monotone under foreground growth), and we report the counterexample, the catch, and the fix as a section of the method, not a footnote. A worst-case-exhaustive soundness harness attacks all four class-valued quantities — coverage, the fifth, is a pixel fraction with no per-instance support — at the two deterministic extremes; on real model outputs it reports **0 violations in 16,675,936 checks**. On 30 real mosaic crops through a frozen zero-shot grout-segmentation model the certificate is non-vacuous: at $\varepsilon = 0.10$, 94% of predicted grout length is certainly-foreground; at the calibration-defined $\varepsilon = 0.40$, 60%. Coverage, however, is not connectivity — the strict certified-connected fraction is always smaller: 0.766 at $\varepsilon = 0.10$ and 0.343 at $\varepsilon = 0.40$, a modest 1.2–1.7× gap on real outputs (the order-of-magnitude gap appears only on the near-tautological toy track, where the blurred maps fragment). We report the two numbers separately rather than blur them into one.

1 Introduction

A segmentation map is usually consumed through its topology. A radiologist asks whether a vessel is interrupted; a conservator asks whether a mosaic's grout network still separates every tile from its neighbours; an inspector asks whether a crack network is one connected system or several. Pixelwise scores answer none of these questions directly, and a model that is 99% accurate per pixel can still be topologically wrong everywhere it matters.

A substantial literature addresses this at *training time*: topology-aware losses and metrics — clDice for tubular structures [24], persistent-homology losses [6, 18], Betti matching [26] — shape the learned map so that its topology is better *on average*. What they do not provide is a guarantee about any given output: after training, nothing tells the consumer which components of *this* prediction survive *this* much uncertainty.

This paper takes the complementary, inference-time view. Given a frozen score map p and an ℓ_∞ budget ε on perturbations of p, we compute topological statements that **hold on every mask reachable within the budget**, and we prove and then *test* that claim adversarially. The name is the guarantee: INTACT certifies what remains *intact* — which components stay connected, which lines stay unbroken, which holes stay unfillable — under any in-budget perturbation. The conceptual ancestor is fifty years old: Benson's algorithm for unconditional life in the game of Go [4] certifies a group alive under *any* sequence of opponent moves via a monotone fixpoint — an adversarial certificate, not a statistical one. Our certificate is the same move for segmentation topology: a monotone construction (thresholding an interval-bounded map) whose conclusions hold against the entire perturbation band.

Contributions, each with its qualifier:

1. **The sandwich is exact** (§2). CERT_FG/POSSIBLE/CERT_BG partitions the image so that the reachable set of binary masks under any ℓ_∞-ε perturbation is *exactly*

$$\{M : \mathrm{CERT\_FG} \subseteq M \subseteq \neg\mathrm{CERT\_BG}\}$$

(Proposition 1) — thresholding is monotone and interval propagation through it is endpoint-exact, so the abstraction loses nothing.

2. **Two masks suffice, and both are needed** (§3.6). Over that interval a finite conjunction of monotone predicates is certified by checking exactly the two extreme masks (Lemma 1); each extreme is individually necessary, and no fixed budget of random masks substitutes for either — the image-topology instance of a classical three-valued duality (§6), made exact and tight.

3. **Five certified quantities, each with adversary and caveat** (§3): certainly-foreground coverage; certified β_0 components; certified enclosed tesserae; a strict certified-connectivity fraction; and certified 1-cycles as *unfillable holes*, reported as the count of enclosed guaranteed-background regions and cross-checked against the rank of the image $H_1(\mathrm{CERT\_FG} \hookrightarrow \neg\mathrm{CERT\_BG})$ — two related quantities that are not equal in general (§3.5). We keep coverage and connectivity as two separate reported numbers, because they are different quantities — a modest 1.2–1.7× apart on real outputs, an order of magnitude only on the near-tautological toy track (§5.2).

4. **Adversarial self-verification as part of the method** (§4). A soundness harness exercises every certified class at the deterministic worst-case extremes of the reachable set (plus random reachable masks) and must also catch deliberately planted unsound certificates. An adversarial-verification pass over the registered claims caught a real unsoundness that the harness, as then configured, had missed — our originally registered cycle certificate, $\beta_1(\mathrm{CERT\_FG})$, is *false* as a robustness claim — before any real-model number was reported. We present the counterexample, why the original check missed it, and the corrected certificate.

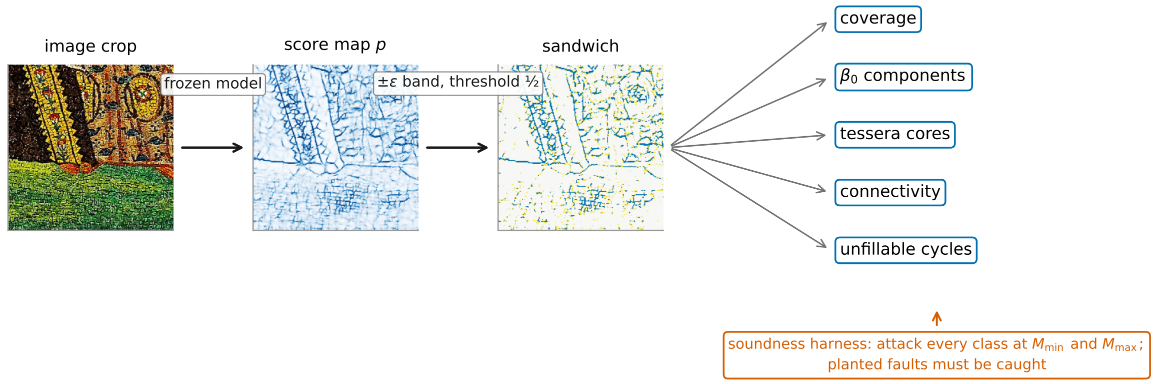

Figure 2: The INTACT pipeline. A frozen segmenter produces the score map p; the $\pm\varepsilon$ band thresholded at $\frac{1}{2}$ induces the sandwich (CERT_FG / POSSIBLE / CERT_BG); five certified quantities are read off, four of which (components, tessera cores, cycles, connectivity segments) carry a support the soundness harness attacks at the two adversarial extremes $M_{\min}, M_{\max}$ (coverage is sound by CERT_FG-containment, not attacked), with planted-fault regressions required to ring.

5. **Non-vacuity on real model outputs** (§5). On 30 real mosaic crops through a frozen zero-shot grout segmenter, the certificate holds with 0 violations in 16,675,936 checks and is far from empty: 94% of predicted grout length is certainly-foreground at $\varepsilon = 0.10$, and 60% survives even at the calibration-defined budget $\varepsilon = 0.40$, with dozens of certified components and hundreds of certified enclosed tesserae; the strict certified-connectivity fraction, now measured on real p, is 0.766 and 0.343 at those two budgets.

Threat-model statement (read first). The certificate as built operates on perturbations of the *score map* p: the adversary may replace p by any p' with $\|p' - p\|_\infty \le \varepsilon$. It is **not** an end-to-end input-space certificate — nothing here bounds how far p can move when the *input image* is perturbed. The two compose cleanly in principle: any input-space method that yields per-pixel bounds on the score map (randomized smoothing [7, 12], interval bound propagation [16]) supplies exactly the interval field our machinery consumes. We state this as an interface in §8; we have not built it, and no claim in this paper depends on it.

Domain. Our primary testbed is GROUT [20], a zero-shot (synthetic-only trained) EfficientNet-B3 U-Net segmentation model for the grout lines of stone mosaics; the enclosed tiles are *tesserae*. Nothing in §§2–4 is specific to mosaics — the machinery consumes any soft map thresholded at $\frac{1}{2}$ — and we exercise that generality off-domain in §5.7 with a cross-domain stress-test on two retinal-vessel datasets (DRIVE [25] and CHASE_DB1 [15], model-free Frangi maps), where the soundness harness holds across a second modality. The *headline non-vacuity* numbers, however, are about the single mosaic task and one frozen checkpoint, and we say so wherever it matters.

2 Setting and threat model

Let $p \in [0,1]^{H\times W}$ be a frozen score map ("grout probability" per pixel), binarised at threshold $\frac{1}{2}$: the predicted foreground is $\{p > \frac{1}{2}\}$ (pipeline overview: Figure 2). The adversary picks any p' with $\|p' - p\|_\infty \le \varepsilon$ and the consumer sees the mask $M(p') = \{p' > \frac{1}{2}\}$.

The sandwich.

$$\begin{aligned}
\text{CERT_FG} &= \{\, x : p(x) - \varepsilon > \tfrac{1}{2} \,\} \quad \text{(certainly foreground)}\\
\text{CERT_BG} &= \{\, x : p(x) + \varepsilon \le \tfrac{1}{2} \,\} \quad \text{(certainly background)}\\
\text{POSSIBLE} &= \text{the complement} \quad \text{(the adversary's territory)}
\end{aligned}$$

The asymmetry between the two tests is forced by the strict threshold rule $M = \{p' > \frac{1}{2}\}$, and is not a blemish: a pixel is certainly foreground iff its lower endpoint clears the threshold strictly, and certainly background iff its upper endpoint fails to — which is the *non-strict* $p + \varepsilon \le \frac{1}{2}$. With a strict test on the background side, a pixel at the exact tie $p + \varepsilon = \frac{1}{2}$ would be called POSSIBLE despite being unable to cross the threshold, and the proposition below would hold only up to such ties.

Proposition 1 (reachable set, exact)**.** $\{\, M(p') : \|p' - p\|_\infty \le \varepsilon \,\} = \{\, M : \text{CERT\_FG} \subseteq M \subseteq \neg\text{CERT\_BG} \,\}$.

Proof. ($\subseteq$) For any admissible p' and any $x \in \text{CERT\_FG}$, $p'(x) \ge p(x) - \varepsilon > \frac{1}{2}$, so $x \in M(p')$; symmetrically no $x \in \text{CERT\_BG}$ enters $M(p')$. ($\supseteq$) Given any M with $\text{CERT\_FG} \subseteq M \subseteq \neg\text{CERT\_BG}$, set $p'(x) = \min(p(x) + \varepsilon,\, 1)$ where $x \in M$ and $p'(x) = \max(p(x) - \varepsilon,\, 0)$ otherwise. Then p' is admissible, and exactly two implications are needed: if $x \in M$ then $x \notin \text{CERT\_BG}$, so $p(x) + \varepsilon > \frac{1}{2}$ and hence $p'(x) > \frac{1}{2}$; if $x \notin M$ then $x \notin \text{CERT\_FG}$, so $p(x) - \varepsilon \le \frac{1}{2}$ and hence $p'(x) \le \frac{1}{2}$. Therefore $M(p') = M$. □

Floating point. In the implementation the interval $[p - \varepsilon,\, p + \varepsilon]$ is widened outward by one ulp per endpoint (`np.nextafter`; enabled by default) before the comparisons, so rounding can only *shrink* the certified sets, never grow them. One consequence: an exact tie $p + \varepsilon = \frac{1}{2}$ has its endpoint widened above $\frac{1}{2}$ and is recorded as POSSIBLE rather than CERT_BG. The computed sets are therefore the exact sets of Proposition 1 minus a measure-zero tie set, always in the direction of abstention — the behaviour a safety-critical consumer needs, since a pixel the budget could push either way is treated as uncertain rather than silently certified.

Thresholding is a monotone operation and interval propagation through it is endpoint-exact: the propagated interval is attained, which is what Proposition 1 states set-theoretically. In abstract-interpretation terms [10]: the interval abstraction of the threshold operator is *exact* here, not merely sound.

Everything certified below is a statement quantified over the reachable set of Proposition 1 — equivalently, over every admissible perturbation.

Structured adversaries. Any perturbation class contained in the ℓ_∞ ball inherits soundness by containment: every certificate holds a fortiori against smooth fields, low-frequency noise, or any other structured subset of the ball. What is surrendered is exactness — the reachable set of a structured class is generally not a lattice interval, so Proposition 1's equality becomes containment and the two-extreme check certifies against a superset of what is strictly reachable: sound but conservative, with the conservatism concentrated at POSSIBLE features smaller than the adversary's correlation length. The tightness measure argument of §3.6 says why this direction is the right one to be conservative in: correlated noise makes coordinated c-pixel flips *likely* rather than measure 2^{-c}, so a structured adversary is more dangerous per unit budget, never less. Exact certification under structured classes is open (§8).

Where ε comes from. So that ε is not an invented number, the "useful" budget is calibration-defined: the median of the confidence margin $|p - \frac{1}{2}|$ over uncertain pixels ($0 < \text{margin} < \frac{1}{2}$), per map, median across maps, snapped to the *nearest* point of the swept grid. On the real track this gives $0.425 \to \varepsilon = 0.40$; the sweep is $\varepsilon \in \{0.02, 0.05, 0.10, 0.15, 0.25, 0.40\}$. Snapping to the nearest point can move ε down, which weakens the adversary and so makes certificates easier — the one step in the pipeline that does not err toward abstention. It is guarded by the split-half stability of the estimator (§5.6), which selects the same grid point in 200 of 200 resamples, and by the exactly-nested sweep (§5.4), which shows what any mis-estimate would cost.

Conventions (part of the certificate's definition, not post-hoc filters). Certificates take a region-of-interest mask and a physical scale floor. The ROI is the whole frame on the mosaic tracks — so the abstain denominator there is the whole image — and, on the retinal tracks, the dataset's field-of-view mask eroded by six binary-erosion iterations. The floor is a *local-thickness* test rather than an area or diameter test: a component survives iff it is non-empty after a morphological opening with an elliptical structuring element of radius $\max(1, \text{round}(\text{floor}/2))$ px, i.e. iff it contains an inscribed disc of that radius. On the mosaic tracks the floor is half the tessera-width scale, tessera width being the square root of the median area of the 4-connected background components of the *ε-independent* prediction $\{p > \frac{1}{2}\}$ (components of area ≥ 9 px; fallback width 6.0 px, hence a 3.0 px floor); on the retinal tracks it is a fixed 1.0 px, since vessel calibre rather than tessera size sets the scale. Foreground uses 8-connectivity, background 4-connectivity (the standard dual pairing, so that curves separate). All

Table 1: The five certified quantities. Every claim is quantified over every reachable mask (Proposition 1).

Quantity	Certified claim	Worst-case adversary	Caveat
1. Coverage (certainly-fg fraction of grout length)	every CERT_FG pixel is foreground in every reachable mask	per-pixel ℓ_∞-ε	a **coverage** statement, NOT connectivity; a covered line pinched through POSSIBLE can still split; fraction + absolute length
2. Certified β_0 components (CERT_FG, 8-conn)	each is connected in every reachable mask	any reachable mask (all are supersets)	CERT_FG's *own* components (absolute count), not the prediction's
3. Certified enclosed tesserae (CERT_BG, 4-conn, ROI + scale floor)	each stays fully background in every reachable mask	any reachable mask	certifies a tessera **core**, not the whole tile
4. Certified-connected grout fraction (segments entirely in CERT_FG)	segment endpoints stay connected under any ε-perturbation	all POSSIBLE $\to$ background	strictly $\leq$ coverage; one POSSIBLE pinch disqualifies; modest on real outputs (1.2–1.7×), order-of-magnitude on toy
5. Certified 1-cycles, two ways: **enclosed regions** (CERT_BG walled by CERT_FG, ROI + floor; the headline) and the **image rank** of H_1 (§3.5)	each enclosed region is an unfillable, unleakable hole in every reachable mask; the rank also lower-bounds $\beta_1(M)$ for every reachable M	both deterministic extremes	**not** equal in general (§3.5): the rank is smaller when one cavity holds several CERT_BG components split by POSSIBLE, larger where enclosure applies its ROI/floor. Only the rank bounds β_1; only enclosure localises

metrics are reported as absolute counts alongside any fraction; ratios never stand alone. The proofs are per-pixel or lattice-theoretic throughout, so Proposition 1 and the coverage, component, tessera-core and per-segment connectivity certificates carry to a cubical grid of any dimension, with dimension entering only through the choice of dual adjacency pairing (in 3D, 26/6 or 18/6 rather than 8/4). The cycle certificate is the one exception, and we flag it rather than gloss it: its persistence form generalises degreewise — the image-rank construction applies to H_k for any k — but its *enclosure* form does not. By Alexander duality, background components that cannot reach the border realise H_{d-1}, which is H_1 only when $d = 2$; in three dimensions the same computation counts enclosed voids, and H_1 tunnels have no enclosure form at all (§3.7, §8).

What is *not* claimed. (i) Nothing about the input$\to$ p pipeline (threat-model statement, §1). (ii) Nothing about correctness of p against ground truth: the certificate concerns robustness of *this map's* topology to ε, not whether that topology is right. (iii) Nothing about POSSIBLE pixels — they are reported as the abstain fraction. (iv) No claim that "the whole grout network certainly stays connected"; the connectivity certificate is per-segment (§3.4).

3 Certified quantities and their soundness

Table 1 itemises the five certified quantities. Each row is a worst-case claim over the entire reachable set, and each carries the caveat that bounds what it means. This table is the paper's claim surface: nothing stronger is asserted anywhere.

3.1 Coverage (certainly-foreground fraction)

The fraction of the skeleton of the predicted foreground that lies in CERT_FG. We report it alongside its absolute certified length in pixels, so the fraction never stands alone (Table 2 caption). Sound by Proposition 1. The point bears repeating because it is the most tempting overclaim in this design space: high coverage does **not** mean the structure stays connected (§3.4, §5.2).

3.2 Certified β_0 components

The connected components of CERT_FG itself (8-connectivity). Every reachable mask M satisfies CERT_FG $\subseteq M$, and adding pixels never disconnects an already-connected set, so each CERT_FG component remains connected — as a set of foreground pixels — in every M. The count is absolute and refers to CERT_FG's own components; it is *not* the number of components of the prediction $\{p > \frac{1}{2}\}$ (at large ε, CERT_FG fragments and this count *rises* — visible in Table 2 — precisely because it counts certified fragments). A scope note a careful reader will want: this is a certificate about the *named* components (each stays connected in every reachable mask), not a lower bound on β_0 of every reachable mask — two certified components can merge through POSSIBLE, which is exactly why the count rises with ε. The quantity that would bound distinct structures from below — the rank of the image $H_0(\text{CERT\_FG} \to \neg\text{CERT\_BG})$, the degree-0 analogue of §3.5's cycle certificate — is well-defined in the same filtration; we leave its registration as a certified quantity to future work.

3.3 Certified enclosed tesserae

The connected components of CERT_BG (4-connectivity), after ROI and scale floor. Each lies in the background of every reachable mask. This certifies the tessera *core* — the region that certainly remains background — not the full extent of the tile.

3.4 Certified connectivity (the strict number)

Skeletonise the predicted foreground within the ROI by Zhang–Suen thinning (`skimage.morphology.skeletonize`) — the thinning rule fixes the node degrees and hence the segment decomposition, so it is part of the certificate's definition — and reduce it to a segment graph: nodes are junctions (degree $\geq$ 3) and endpoints (degree 1); edges are the junction-to-junction skeleton segments. A segment is **certified-connected** iff *every* pixel of it lies in CERT_FG; then its endpoints remain connected under any admissible perturbation, because the whole segment is present in every reachable mask. A single POSSIBLE pixel on the segment disqualifies it — that pixel is the adversary's, and the worst-case mask (all POSSIBLE $\to$background) cuts the segment there.

We report the certified fraction of segments and of skeleton length (the absolute segment and length companions are in the Table 2 caption). By construction this is bounded above by coverage; that gap is measured, not hidden — modest on real p (1.2–1.7×) and dramatic only on the toy track (§5.2). We explicitly do not claim network-level connectivity — only the per-segment guarantee.

3.5 Certified 1-cycles (unfillable holes) — the corrected certificate

A cycle of the prediction is worth certifying only if the adversary can neither *fill* it (turn its interior to foreground) nor *leak* it (break its wall). The certified object is therefore an **unfillable hole**: a connected region of CERT_BG (guaranteed background in every reachable mask) enclosed by CERT_FG (guaranteed foreground). Two related computations, which are *not* equivalent in general:

- **Enclosure form:** CERT_BG components not reachable from the image border through ¬CERT_FG (the can-be-background region); this form is ROI/scale-floor filtered and provides localisation.
- **Persistence form:** the rank of the image $H_1(\text{CERT\_FG} \hookrightarrow \neg\text{CERT\_BG})$, computed by cubical persistent homology [9] with the three-level filtration CERT_FG = 0, POSSIBLE = 1, CERT_BG = 2 (gudhi [21]; intervals with birth < 0.5 and death > 1.5).

The two forms count different things, and we separate them rather than identify them. The enclosure form counts *enclosed guaranteed-background regions*; the persistence form counts the *rank* of an inclusion-induced map. They diverge whenever one enclosed cavity contains several CERT_BG components split

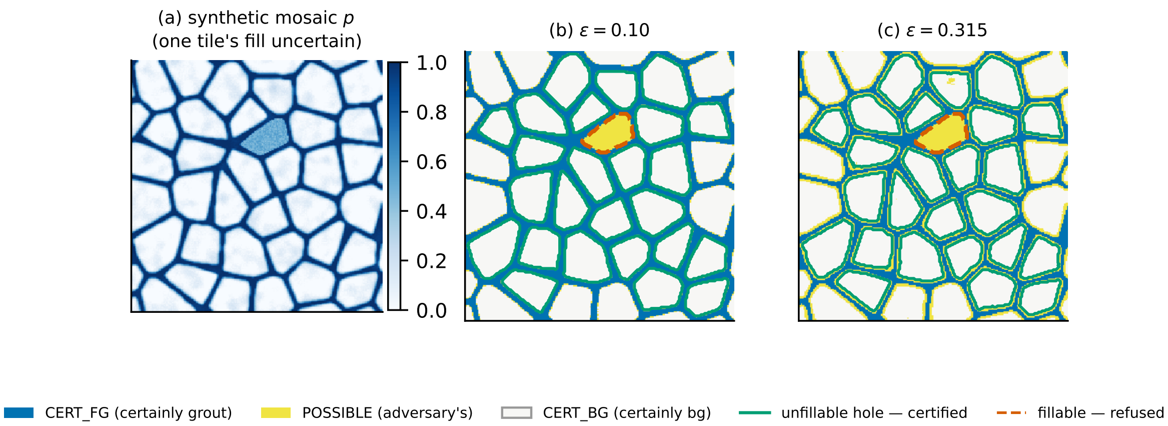

Figure 3: The certified-cycle machine on a synthetic Voronoi mosaic (the interactive companion's view, recreated for print). Confident tiles are certified unfillable holes (green): CERT_BG cores walled by CERT_FG. One tile's interior is set model-uncertain to illustrate the distinction — its enclosed region holds no CERT_BG core, so it is a fillable hole and is *refused* (dashed), exactly what the unsound β_1(CERT_FG) of §4 would still have certified. (b) $\varepsilon = 0.10$; (c) $\varepsilon = 0.315$ (off the swept grid, chosen to show the erosion): the thin joints erode to POSSIBLE while the certified holes and the refusal both persist.

by POSSIBLE: a CERT_FG annulus whose interior holds two CERT_BG squares separated by a POSSIBLE channel gives enclosure 2 but image rank 1 — $\beta_1(\text{CERT\_FG}) = 1$ while $\beta_1(\neg\text{CERT\_BG}) = 2$, and the annulus generator maps to the *sum* of the two blob classes. Only the persistence rank lower-bounds $\beta_1(M)$ for every reachable M (at M = CERT_FG the two squares merge into a single hole). The enclosure count stays sound as a *per-region* statement — each region is guaranteed background and unleakable in every reachable mask, which is what the soundness harness verifies — but it is not a Betti number and must not be read as one. This is the degree-1 instance of the named-features-versus-rank-bound distinction drawn for β_0 earlier in this section.

The two counts agree **exactly** on the clean regression cases (ring-over-POSSIBLE $\to$ 0; ring-over-CERT_BG $\to$ 1; a 3×3 grid of enclosed cells $\to$ 9; the planted unsound cycle $\to$ not certified) — we pre-registered this equivalence as a prediction before the persistence code ran, and it holds wherever each enclosed cavity contains a single CERT_BG component. On noisy maps the two counts separate in both directions — persistence above enclosure where enclosure applies its ROI and scale floor, enclosure above persistence where a POSSIBLE channel subdivides one cavity — and §5.5 gives the measured decomposition. Localisation stays with the enclosure form: persistence gives a certified *count*, not per-cycle generators.

This definition is the *corrected* one. It is not what we first built — §4 reports what we first built, why it was wrong, and how it was caught.

3.6 Soundness: the structural argument and the adversarial check

Soundness has two layers, and we state the strength of each precisely.

The structural layer. Each certificate is sound by construction on top of Proposition 1: CERT_FG is contained in the foreground of every reachable mask (quantities 1, 4); connectivity is preserved under superset (quantity 2); CERT_BG is contained in the background of every reachable mask (quantity 3); and an unfillable hole's wall and interior are pinned by both containments at once (quantity 5). The outward one-ulp widening of §2 only shrinks CERT sets — the safe direction — and the strict inequalities settle exact boundary ties conservatively.

The adversarial layer. An implementation can betray a proof, so every certified class emitted by the pipeline is attacked by an explicit soundness check: the *deterministic worst-case extremes* of the reachable set — all POSSIBLE pixels to background (the minimal mask, the worst case for anything foreground) and all POSSIBLE pixels to foreground (the maximal mask, the worst case for anything background or for filling a hole) — plus random reachable masks. That the two extremes suffice is not an empirical observation; it is a consequence of the predicates' monotonicity:

Lemma 1 (two extremes are exhaustive)**.** *Write the reachable set as the lattice interval*

$$\mathbb{I} = \{M : M_{\min} \subseteq M \subseteq M_{\max}\}, \qquad M_{\min} = \text{CERT\_FG}, \quad M_{\max} = \neg\text{CERT\_BG}$$

(Proposition 1). Call a predicate $\varphi(M)$ upward monotone *if $\varphi(M)$ and $M \subseteq M'$ imply $\varphi(M')$,* downward monotone *dually. Then a finite conjunction of monotone predicates holds on all of $\mathbb{I}$ iff every upward-monotone conjunct holds at $M_{\min}$ and every downward-monotone conjunct holds at $M_{\max}$.*

Proof. An upward-monotone conjunct that holds at $M_{\min}$ holds at every $M \supseteq M_{\min}$; if it fails, $M_{\min} \in \mathbb{I}$ is itself the witness. Dually at $M_{\max}$. A conjunction fails somewhere on $\mathbb{I}$ iff some conjunct does. □

Each harness predicate is such a conjunction: a foreground class asserts $S \subseteq M$ for its (already-connected) support — upward monotone, worst case $M_{\min}$; a background core asserts $S \cap M = \emptyset$ — downward monotone, worst case $M_{\max}$; a cycle asserts its wall is present (upward, $M_{\min}$) *and* its interior stays background (downward, $M_{\max}$) — both extremes together; a connectivity segment asserts a path exists — path existence is upward monotone, worst case $M_{\min}$. A violation counts as a broken certificate; a single one halts the run before any headline number is emitted.

Tightness (both extremes necessary). Lemma 1 makes the pair *sufficient*; the certified classes make each extreme *individually necessary* (Figure 4). Drop $M_{\max}$ and Figure 5's fillable ring is falsely certified — its cycle holds at $M_{\min}$ (wall present, interior open) yet is filled at $M_{\max}$ once the POSSIBLE interior is forced to foreground (§3.5); drop $M_{\min}$ and Figure 8's pinched bar is falsely certified — its segment spans at $M_{\max}$ yet is cut at $M_{\min}$ once the POSSIBLE pinch is forced to background (§3.4). By Lemma 1 each failure localises at exactly one extreme, never strictly interior; the foreground-support and background-core classes only fix *which* extreme is worst ($M_{\min}$ and $M_{\max}$ respectively, the direction in which the harness would reject a support or core straying into POSSIBLE), and a cycle forces both at once. Since $M_{\min}$ is the worst case of every upward-monotone conjunct and $M_{\max}$ of every downward one, neither extreme dominates the other: neither can stand in for the other, and the check is *minimal* at two masks — each extreme is itself a reachable mask on which the dropped-extreme harness fails, so omitting either forfeits worst-case certainty. Random sampling cannot restore it: by monotonicity $M_{\min}$ lies in *every* non-empty upward-violation region and $M_{\max}$ in every downward one, whereas a random draw detects a violation only by landing in that region, of measure 2^{-c} in the number c of POSSIBLE pixels that must turn adversarial in unison (an enclosure's whole interior, a segment's separating cut). Taking c large — an enclosure of growing area — sends the per-draw detection 2^{-c} to zero, so for any fixed budget N the miss probability $(1-2^{-c})^N \to 1$; the two extremes drive all POSSIBLE pixels at once and catch it with certainty. The random reachable masks therefore add nothing to worst-case coverage; they are retained only as a cross-check against implementation faults (an independent review of the harness reached the same exhaustiveness conclusion before Lemma 1 was written down). The measure argument is itself measured: a registered baseline (Appendix C, Figure 12) plants fillable rings and pinched bars with controlled in-unison count c, scores the semantic events, and finds per-draw detection matching 2^{-c} exactly under the uniform model — polynomially better yet still vanishing under the harness's own sampler — with every planted violation caught at exactly the predicted extreme and never the other.

The harness must also be able to *fail*: deliberately planted unsound certificates — a foreground class intruding into POSSIBLE, a background core intruding into POSSIBLE, a "certified" cycle over a fillable interior, a "certified-connected" segment spanning a pinch — are all flagged by the same check (regression tests). A certificate system whose alarm never rings is indistinguishable from one with no alarm; ours rings on every planted fault. The one real fault (§4) was caught not by this harness — which did not yet know cycles were a class — but by an external adversarial-verification pass, after which the harness was extended so that it would ring.

Result. 0 violations for every certified class emitted on each track and all swept ε — all four classes (components, tessera cores, corrected enclosure cycles, and connectivity segments) on both the toy track and the real track (16,675,936 checks over 30 crops).

3.7 Generality of the construction

Interval fields (Proposition 1′). Nothing in Proposition 1 uses the uniformity of ε: the proof is per-pixel, so for any per-pixel interval field $[l(x), u(x)]$ on the score map the reachable set of masks is exactly the lattice interval between $\{l > \frac{1}{2}\}$ and the complement of $\{u \leq \frac{1}{2}\}$, and every certificate above

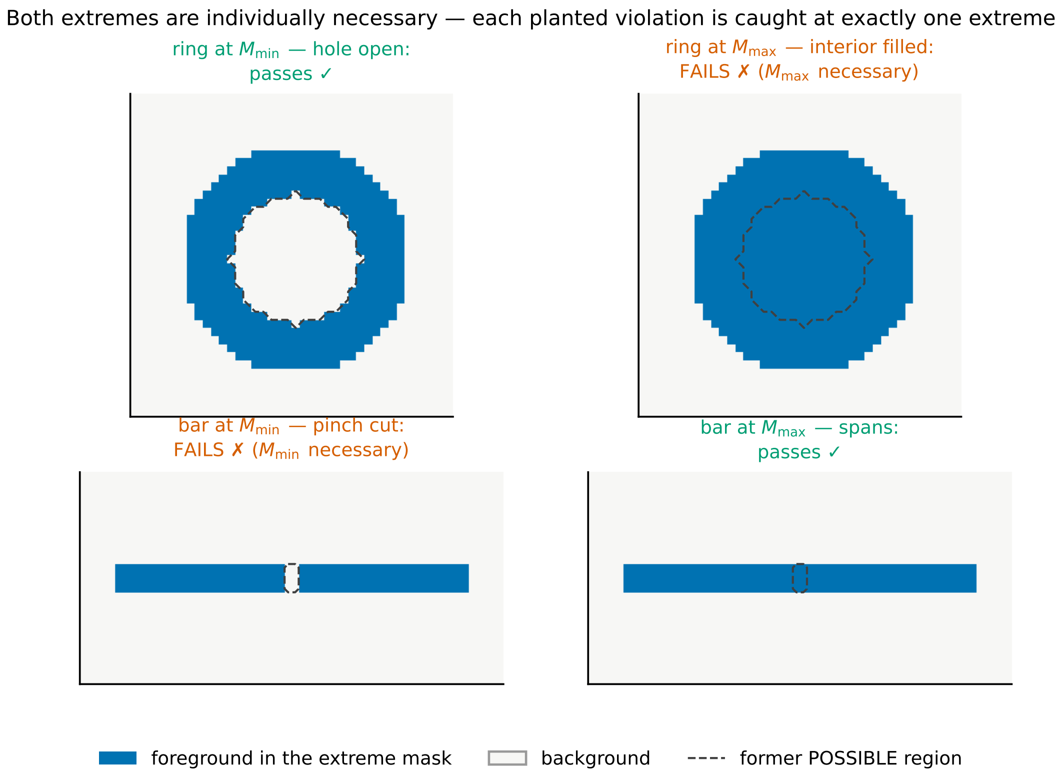

Figure 4: Tightness of the two-extreme check: the fillable ring's cycle certificate holds at $M_{\min}$ (hole open) and fails only at $M_{\max}$ (interior filled), so dropping $M_{\max}$ falsely certifies it; the pinched bar's segment spans at $M_{\max}$ and fails only at $M_{\min}$ (pinch cut), so dropping $M_{\min}$ falsely certifies it. Dashed outline: the POSSIBLE region the adversary controls. By Lemma 1 each failure localises at exactly one extreme, never strictly interior.

applies unchanged to the resulting (non-uniform) sandwich. Uniform ε is the corollary $l = p-\varepsilon$, $u = p+\varepsilon$, and the composition interface of §8 is the corollary for bounds supplied by input-space certification. The implementation is already field-ready — a per-pixel ε array broadcasts through the identical code path (regression-tested) — though the harness results of §5 exercise only the uniform case.

The lattice form, and what else is monotone. More abstractly, the ingredients are an interval in a complete lattice and monotone predicates: the sandwich is a three-valued (Kleene) abstraction and the two-extreme check is the classical optimistic/pessimistic-completion duality (§6) — we claim the imaging instantiation, its exactness (Proposition 1), and its tightness (§3.6), not the lattice principle. The same two-mask check therefore extends beyond topology: morphological erosion and dilation are monotone lattice maps, so certified minimum thickness (survival of erosion at $M_{\min}$), certified clearance (dually at $M_{\max}$), and certified area intervals follow with no new theory — geometric certificates we flag but do not build here.

What is not monotone. The recipe has a boundary, and §4's correction is its instance: perimeter, skeleton length, convexity (which fails in both directions), curvature, and the Euler characteristic are *not* monotone under foreground growth — and β_1 fails directly, since growth fills holes, which is exactly why the naive cycle certificate was unsound and the corrected one decomposes into a wall conjunct (upward) and an interior conjunct (downward), one per extreme. Non-monotone quantities require such a decomposition or a different certificate; Lemma 1 as stated does not cover them, and we do not claim it does.

4 The correction: an unsound cycle certificate, caught and fixed

We consider this section part of the method. Adversarial self-verification is not an afterthought to the certificate; it is what makes the certificate credible — a certificate system that cannot catch a planted

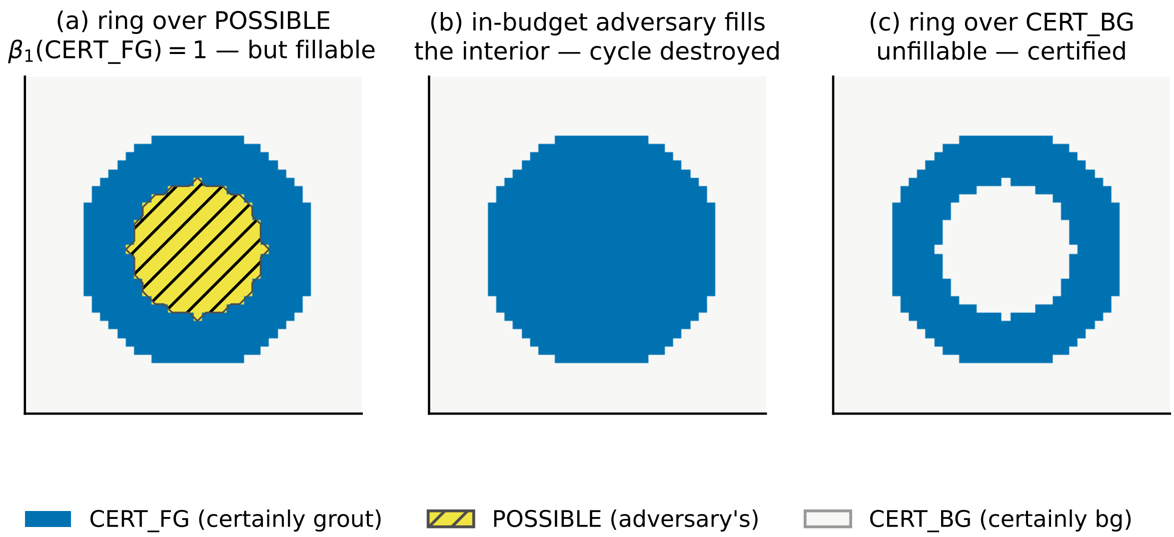

Figure 5: The β_1 counterexample (regression-test constructions, $\varepsilon = 0.1$). (a) A CERT_FG ring over a POSSIBLE interior: $\beta_1(\text{CERT\_FG}) = 1$, but (b) the all-POSSIBLE →foreground extreme — a reachable mask — fills the hole; certifying this cycle was unsound. (c) The corrected certificate requires the interior to be CERT_BG: an unfillable hole.

unsound certificate should not be trusted, and one that never had to correct itself has probably not been attacked hard enough.

4.1 What we originally certified

As first registered, certified 1-cycles were computed as $\beta_1(\text{CERT\_FG})$, with the justification: "a loop in the minimal foreground persists in all larger foregrounds."

4.2 The counterexample

That justification is **false**: H_1 is not monotone under foreground growth. Adding pixels can only merge or preserve components (which is why quantity 2 *is* sound), but it can *destroy* holes by filling them. Concretely (Figure 5): let CERT_FG be a ring whose interior is POSSIBLE. Then $\beta_1(\text{CERT\_FG}) = 1$, and the original certificate called this cycle robust. But the adversary may, within budget, push every interior pixel above threshold — a reachable mask by Proposition 1 — and the hole is gone: $\beta_1(M) = 0$. An in-budget perturbation destroys the "certified" cycle. The certificate was unsound. (In persistence terms the trap is quantitative: this ring's superlevel bar has persistence 0.45 — comfortably above $2\varepsilon = 0.2$ — yet the hole dies, because a long bar need not straddle the threshold window at $\frac{1}{2}$; see §6.)

4.3 Why the in-code check missed it

The soundness harness of §3.6 existed at the time and passed — because it tested only the component and tessera classes. Cycles were never attacked: the check enumerated worst-case masks for the classes it knew about, and nobody had told it a cycle was a class. The lesson is specific: *every* certified predicate needs its own adversary in the harness, and a predicate without one should be treated as unverified.

4.4 The fix

The corrected certificate is the unfillable hole of §3.5: a cycle is certified iff its enclosed region is CERT_BG (so it cannot be filled) walled by CERT_FG (so it cannot leak). The harness gained a `cycle` class attacked at both extremes, and the regression suite now encodes the exact counterexample: ring-over-POSSIBLE is *not* certified; ring-over-CERT_BG is certified and survives the attack; a planted unsound cycle is caught. The equivalence of the enclosure computation with cubical image-persistence (§3.5) was registered and verified afterwards as an independent cross-check on the fixed definition.

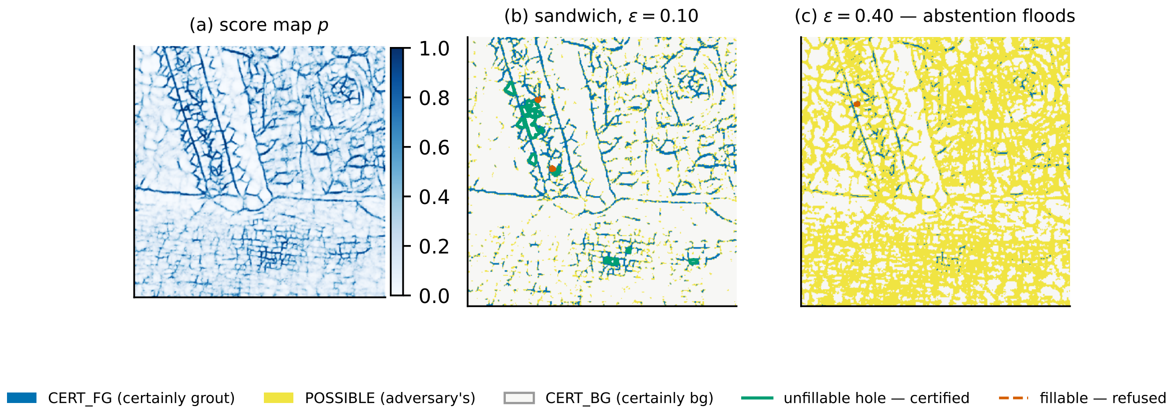

Figure 6: Anatomy of the certificate on a real crop — the *worst-coverage* crop at $\varepsilon = 0.10$ by the registered metric (`crop_byzantine_01`), not a curated example. (a) frozen-model score map; (b) the sandwich at $\varepsilon = 0.10$ — CERT_FG grout (blue), CERT_BG tesserae (grey), POSSIBLE (yellow) — with certified unfillable holes outlined in green; (c) at the calibration-defined $\varepsilon = 0.40$ the abstention band floods and fewer classes survive: certification degrades monotonically (§5.4), never silently.

4.5 Provenance

The unsoundness was found on 2026-07-12 by an adversarial-verification pass (two independent AI soundness reviewers attacking the registered claims; the review record is preserved with the released code), *before* the real-model track ran: no real-p number was ever reported under the unsound definition. In the same pass, a metric-semantics overclaim was corrected: a "certified component fraction" (predicted components $\geq$50%-covered by CERT_FG) was renamed to a coverage proxy, because a $\geq$50%-covered component pinched through POSSIBLE can still split — coverage is not connectivity, the recurring theme of this paper. The registry preserves the original claims verbatim alongside the correction addendum; the paper you are reading states only the corrected forms.

5 Experiments

Setup. Real track (T-real): 30 real mosaic crops (the full frozen evaluation set) through a frozen zero-shot GROUT checkpoint (`grout_b3_zeroshot_v1`) [20]; p is the sigmoid of the model's patch logits. Crops are 512×512 px, the unit the runtimes of Appendix E are quoted per. Toy track (T-toy): 24 synthetic 256×256 mosaic masks from our own generator with simulated scores

$$p = \mathrm{clip}\big(\mathrm{GaussianBlur}(\mathrm{mask}, \sigma{=}1.3) + \mathcal{N}(0, 0.07^2)\big)$$

— retained strictly as a validation track (§5.3). Sweep and conventions as in §2; seeds fixed; the toy run that carries the (persistence cycles, connectivity) is the run reported in Table 3. The regression suite — including the planted-unsound and counterexample cases — passes (§10).

5.1 T-real: sound and non-vacuous on real model outputs

Soundness first, because nothing else matters if it fails: the soundness gate passes with **0 violations in 16,675,936 checks** across all four certified classes this run emitted (components, tessera cores, corrected cycles, and connectivity segments) and all ε (§3.6). The certificate then turns out to be far from vacuous on real outputs (Table 2, Figure 7 left).

Readings. At an illustrative mid-sweep $\varepsilon = 0.10$, **94%** of predicted grout length is certainly-foreground with 20.3 certified components and 439.0 certified enclosed cycles per crop on average — and, now measured on real p, **0.766** of grout length is certified to stay *connected* (unbroken). The model is confident on these crops, so the calibration-defined useful budget is large (0.425, snapped to 0.40); even there **60%** of grout length is certainly-foreground and **0.343** is certified-connected, with 41.7 certified components and 171.4 certified cycles (per-crop means). Worst-case over the 30 crops — the right headline for a worst-case paper — the cohort floors hold: coverage 0.187 and certified connectivity 0.028 at $\varepsilon = 0.40$ (each the minimum over the 30 crops, on *different* crops; Figure 11, Appendix A, shows

Table 2: T-real: certified quantities vs. ε (frozen zero-shot model on 30 real 512×512 crops; per-crop means). No ratio stands alone. Coverage is certified skeleton length over total skeleton length: 23,408.1 and 14,969.8 px at $\varepsilon = 0.10/0.40$, over 24,728.9 px per crop. Conn-seg is 2,173.2 and 1,003.5 certified segments over 2,739.8. Conn-len is certified segment length over TOTAL SEGMENT length, a smaller denominator than the skeleton (node clusters adjacent to no degree-2 interior belong to no segment); its absolute pair is not persisted in the results file. Both skeleton length and segment count are ε-independent. Abstain is over the ROI, here the whole crop. Each fraction is the mean of the per-crop ratios, not the ratio of the means.

eps	coverage	conn-len	conn-seg	#comp	#tess	#cyc-encl	#cyc-pers	abstain
0.02	0.988	0.921	0.916	17.9	530.5	471.5	740.6	0.016
0.05	0.971	0.848	0.841	18.1	533.2	462.9	684.9	0.040
0.10	0.942	0.766	0.761	20.3	537.3	439.0	606.6	0.080
0.15	0.908	0.700	0.696	22.5	540.2	411.4	532.0	0.123
0.25	0.823	0.577	0.575	30.2	536.4	342.7	398.4	0.219
0.40	0.598	0.343	0.348	41.7	474.6	171.4	186.7	0.438

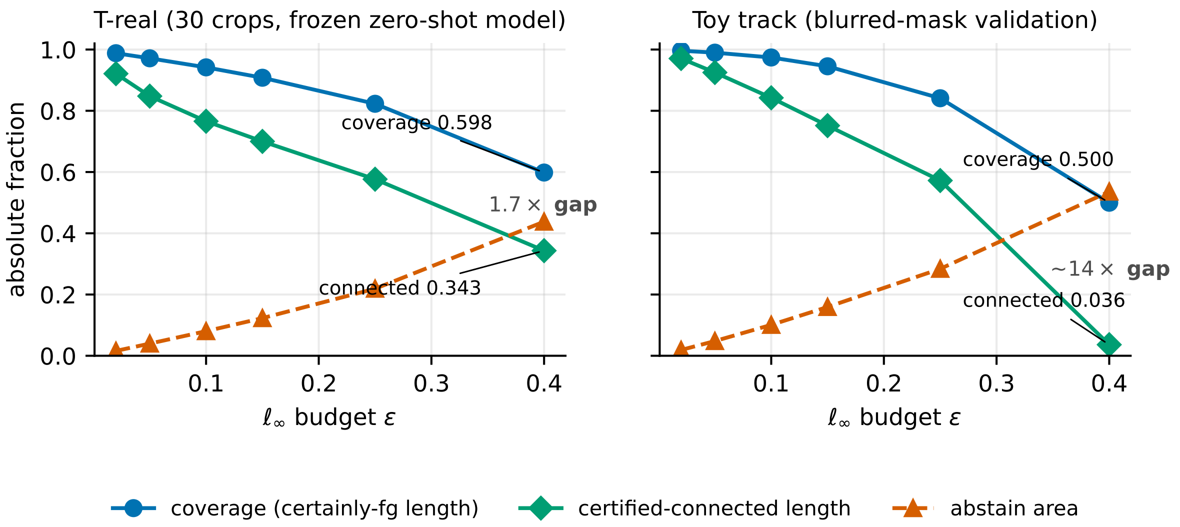

Figure 7: Certified fractions vs. ε, both tracks (coverage, certified-connected length, abstain). Left (T-real, 30 crops): the coverage-vs-connectivity gap is a *modest* 1.7× at $\varepsilon = 0.40$ (0.598 vs 0.343) — the confident model keeps most segments entirely inside CERT_FG. Right (T-toy): the same gap collapses to ∼14× (0.500 vs 0.036), an artifact of the blurred toy maps whose skeletons fragment *en masse*. Same three series, same y-scale; the panels' contrast is the point (§5.2).

the full per-crop distribution). Our pre-registered non-vacuity criterion — at least 50% of grout length certainly-foreground, with at least one certified component, at the useful ε — passes; the pre-registered kill condition (under 10% certified at every useful ε) is not triggered. The component count *rises* with ε (17.9 → 41.7) — that is CERT_FG fragmenting, counted as certified fragments, not evidence of more structure.

5.2 Coverage is not connectivity — the headline pair

The largest number in this paper (94% coverage) is a coverage statement. The corresponding *connectivity* certificate — grout that certainly stays unbroken — is a different, strictly smaller number, and we report both everywhere, never one for the other. We have now measured it on real p: at the useful $\varepsilon = 0.40$, coverage is 0.598 while the strict certified-connected length fraction is **0.343** — a **1.7**× gap (**1.2**× at $\varepsilon = 0.10$). The gap is real — connectivity is always the smaller number — but on real model outputs it is *modest*, because the confident model keeps most segments entirely inside CERT_FG.

The order-of-magnitude gap lives only on the near-tautological toy track, where at $\varepsilon = 0.40$ coverage is 0.500 versus certified-connected length **0.036** (segment fraction 0.062) — a ∼14× collapse (Figure 7 right; Figure 8 shows the mechanism). That collapse is a property of the toy maps, not the certificate: the toy scores are lightly-blurred masks whose skeletons turn POSSIBLE *en masse* at large ε, whereas the real maps are sharp enough that connectivity largely survives (0.343 of grout length certified unbroken

Table 3: Toy track (24 synthetic maps): all columns, including the certified-connected fractions and the persistence cycle counts. Toy p is a blurred copy of the ground truth — validation of the machinery, not evidence about real-model behaviour.

eps	coverage	conn-len	conn-seg	#comp	#tess	#cyc-encl	#cyc-pers	abstain
0.02	0.996	0.971	0.969	0.7	185.8	151.71	214.58	0.019
0.05	0.990	0.925	0.926	0.6	180.4	146.38	203.08	0.048
0.10	0.974	0.842	0.854	0.6	169.4	135.12	183.96	0.101
0.15	0.945	0.752	0.773	0.8	157.0	117.67	152.04	0.159
0.25	0.841	0.572	0.598	3.5	134.6	71.12	73.25	0.284
0.40	0.500	0.036	0.062	18.1	28.2	0.04	1.38	0.536

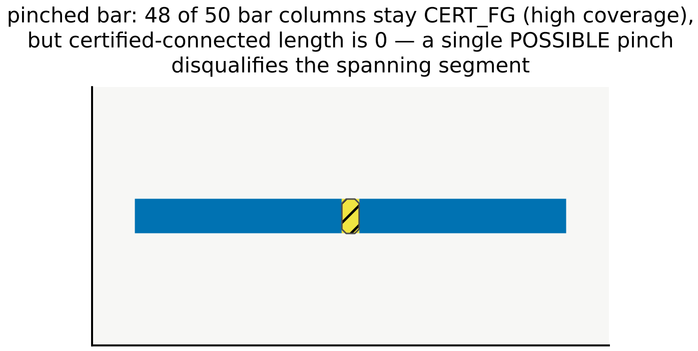

Figure 8: Why coverage $\neq$ connectivity (regression-test construction, $\varepsilon = 0.1$): 48 of 50 bar columns stay CERT_FG (high coverage), but the two-pixel POSSIBLE pinch disqualifies the spanning segment — certified-connected length 0.

even at the largest calibrated budget). Reporting the two numbers separately still matters — blurring them converts a sound coverage statement into an unsound connectivity claim, the same failure mode as §4 at the level of prose rather than code — but the real-track headline is a 1.7× gap, not an order of magnitude.

5.3 T-toy: validation track, with its caveat stated verbatim

The toy scores are GaussianBlur(GT mask, σ=1.3) + noise 0.07 — in the words of the results file: "a lightly-blurred copy of the answer key. Certifying that its topology survives small perturbations is close to tautological, and the 'useful ε' is calibrated from the *same* p, so clearing 50% is partly self-fulfilling. These numbers demonstrate the machinery runs and produces non-empty certificates on plausible-looking input — nothing about real-model behavior." We still print the toy table because engineered maps are where soundness is easiest to attack and where the persistence-cycle (persistence cycles, connectivity) were first validated — not because it is the only track carrying them; the real track now does too (Table 2, §5.1).

On this run's own calibration (0.361 → $\varepsilon = 0.40$) the ≥50%-coverage criterion narrowly fails (0.4998) — reported for completeness; per the caveat above, toy non-vacuity is illustrative either way. Soundness on the toy track: 0 violations, including the connectivity predicate.

5.4 Nesting

Certificates over decreasing ε are monotone-nested — CERT_FG(ε') $\supseteq$ CERT_FG(ε) for $\varepsilon' \le \varepsilon$, exact set containment, so every certified quantity that is a *measure* of the certified sets — coverage, certified length, abstain fraction — is monotone along the sweep. Counts are not, and the tables show it: component and cavity counts are not monotone functionals of set inclusion (§3.7), so #comp and #tess move non-monotonically in Tables 2 and 3 as CERT_FG fragments and CERT_BG merges. One topological count *is* provably monotone: by the diagram identity of §5.5 the persistence-form cycle count at ε is the number of points of p's fixed superlevel diagram straddling $[\frac{1}{2} - \varepsilon, \frac{1}{2} + \varepsilon]$, and widening ε can only shrink that set, so #cyc-pers is non-increasing in ε — on every crop individually, not merely in the mean. Exact set containment is asserted and unit-tested (a pre-registered prediction; it holds on both tracks, and registers the induced monotonicity of the coverage column). This is the hook for the graded extension of §8: the sandwich family over ε *is* a nested (α-graded) structure already.

5.5 Persistence vs enclosure: an equivalence that holds only locally

We pre-registered the two computations of §3.5 as equivalent. That is confirmed on the regression cases (0/1/9 on ring-over-POSSIBLE / ring-over-CERT_BG / 3×3 grid of holes; planted-unsound not certified) and **refuted as a general identity** — we record the refutation rather than quietly narrowing the claim, as with §4. The two columns diverge in *both* directions, for two distinct reasons. Persistence exceeds enclosure where enclosure applies the ROI and scale floor: on noisy toy maps at $\varepsilon = 0.40$, persistence 1.38 vs floored enclosure 0.04 (per-map means), the difference being sub-floor debris the definition excludes. Enclosure exceeds persistence where a POSSIBLE channel subdivides one cavity into several guaranteed-background regions (§3.5); this occurs on real p at the larger budgets. The equivalence therefore holds exactly when each enclosed cavity contains a single CERT_BG component, and not otherwise. The enclosure column remains the headline count — it is the localisable, per-region certificate — and the persistence column is reported alongside it as the rank bound, not as the same number measured twice.

The identity goes further (pre-registered before the check ran): the persistence-form count at *every* swept ε equals the number of points of p's own superlevel diagram straddling the window $[\frac{1}{2} - \varepsilon, \frac{1}{2} + \varepsilon]$ — verified exactly on every crop and every ε (a registered check; §10), so the entire ε-sweep of certified cycles is derivable from one diagram per crop, and §6's rank identity is machine-checked rather than asserted.

5.6 The ε estimator is split-half stable

Because ε is estimated (median confidence margin, §2) rather than chosen, its stability is an empirical question, and we register and measure it: across 200 random 15/15 splits of the real crops, the calibration estimator's split-half drift is median 0.016 (maximum 0.071) — small against the 0.15 gap between adjacent sweep points at the operating budget — and the two halves select the *same* grid point in 200 of 200 resamples, so the held-out certified quantities are identical by construction. We do not *learn* ε against the certified quantities, and would not: the band is the assumption the certificate is conditioned on, and optimizing it against certified counts would invert the semantics (a band chosen to flatter the model certifies nothing about the domain). The estimator is certificate-blind by design — it reads only the score map's margins — and the exactly-nested sweep (§5.4) already quantifies the consequence of any mis-estimate.

5.7 Cross-domain replication: two retinal-vessel datasets

Nothing in §§2–4 is specific to mosaics (§3.7), so we stress-test the certificate off the mosaic domain entirely, on retinal blood-vessel maps — the setting where connectivity is the *clinical* question. We use DRIVE [25] (20 images) and CHASE_DB1 [15] (20 images) and, with **no trained model**, build a score map p per image from multiscale Frangi vesselness [14] under three monotone normalizations (cbrt, sqrt, log) with an Otsu-calibrated logistic, then run the identical certificate.

Soundness holds cross-domain — even on degenerate maps. The headline is negative space: across both datasets and all three normalizations the soundness harness reports **0 violations in 10,578,388 checks** (DRIVE 2,963,152 at the full 16-mask setting; CHASE 7,615,236 at 4 masks — the two deterministic extremes plus two cross-checks, exhaustive by Lemma 1, for the denser vessel maps). This includes a genuinely adversarial regime for the certificate: CHASE's cbrt normalization badly over-segments (median foreground ≈29% of the field of view, per-image Dice against the manual tracing only 0.30; visible as the noisier map in Figure 9), so the certificate must reason about *hundreds* of certified components and thousands of connectivity segments per image — and it stays sound. A worst-case certificate that only held on well-behaved maps would be far less useful; ours holds regardless of the underlying prediction's quality.

Coverage is accuracy-linked; the topological certificates are not simple Dice proxies (Figure 10). Correlating each per-image certified quantity with that image's Dice (Spearman ρ, primary cbrt normalization): coverage tracks Dice on both datasets ($\rho = +0.65$ on DRIVE, $+0.60$ on CHASE,

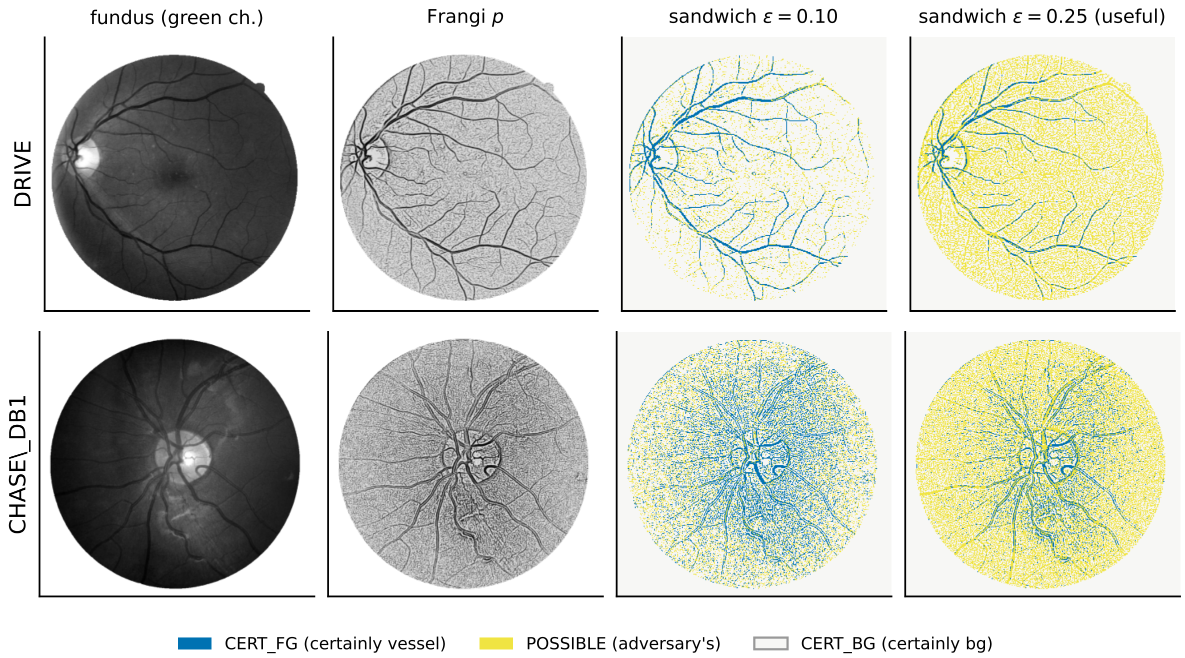

Figure 9: The certificate on real retinal vessels (DRIVE and CHASE_DB1), model-free Frangi score maps, no training or checkpoint. Per row: fundus green channel → Frangi p → the $\pm\varepsilon$ sandwich at a small budget ($\varepsilon = 0.10$, most of the vessel tree certified blue) and at the useful budget ($\varepsilon = 0.25$, the certificate abstains — yellow — over the thinner, noisier regions, more so on CHASE's over-segmented map). The identical harness certifies both domains.

both $p < 0.05$) — expected, since coverage is essentially a confidence measure. The topological invariants behave differently and *inconsistently*: on DRIVE they decouple from Dice (certified-connected fraction $\rho = +0.15$, β_0 $\rho = -0.42$, both n.s.), whereas on CHASE's degenerate cbrt maps they correlate ($\rho = +0.57, +0.51$, both $p < 0.05$) — because in that over-segmented regime more certified structure and less-bad accuracy degrade together; under the better-behaved sqrt/log normalizations they decouple on both datasets. The reading is not "topology is orthogonal to accuracy": it is that the certified topological quantities are **not reducible to Dice**, and their relationship to it is regime-dependent — so they must be reported as their own numbers (the recurring theme of §5.2), never inferred from an accuracy score. The full per-transform ρ table lives in the committed results files.

5.8 An inter-rater budget: ε from annotation disagreement

Where two human tracings exist, ε need not come from the model's own margins (§2) — it can be read from *inter-observer disagreement*, the irreducible uncertainty of the task itself. On the CHASE images with two independent manual tracings, the median confidence margin $|p - \frac{1}{2}|$ over the pixels the two annotators label differently is 0.246, snapping to $\varepsilon = 0.25$ (the two tracings agree at Dice 0.773). Certifying at this budget answers a specifically clinical question — which topological features survive perturbations no larger than the disagreement between two trained human annotators — and grounds ε in a medical protocol rather than a model statistic. The certificate machinery is unchanged.

6 Related work

Certified segmentation via randomized smoothing. Smoothing-based certification [7], scaled to segmentation [12], certifies *pixelwise labels* under input-space perturbations, typically with an abstain mechanism. This is the natural *supplier* for our machinery, not a competitor: pixelwise certified label bounds are an interval field on the score map, which is exactly what the sandwich consumes (§8). What smoothing does not provide — and what we add on the map level — is the topological layer: components, enclosures, connectivity, cycles, quantified against the whole reachable set at once. As of this writing no smoothing-based segmentation certifier — including the hierarchical AdaptiveCertify [2] — certifies any topological invariant; all operate on per-pixel labels.

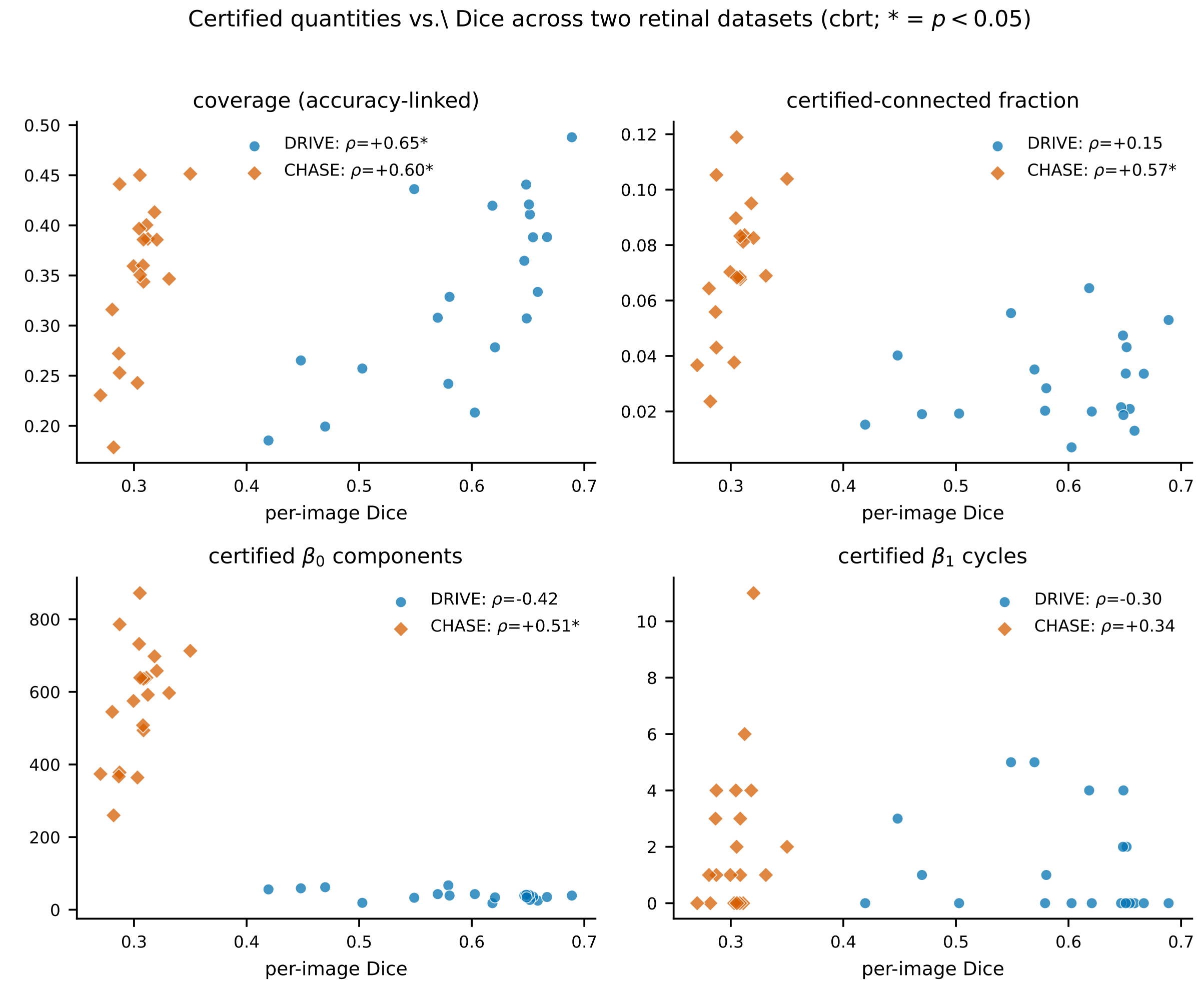

Figure 10: Per-image Dice vs. each certified quantity across DRIVE and CHASE_DB1 (primary cbrt normalisation; Spearman ρ per dataset, $* = p < 0.05$). Coverage tracks Dice on both sets; the topological invariants scatter on DRIVE but slope up on CHASE's degenerate, over-segmented maps (Dice ≈0.30, β_0 counts in the hundreds). Read the two clouds, not one trend line: the same quantity correlates on one dataset and not the other.

Topology-aware training. clDice [24], persistent-homology losses [6, 18], and Betti matching [26] improve topological fidelity of the *learned* map. They are complementary in the strongest sense: they change p, we certify a given p. None of them yields a worst-case statement about a particular output; that is not a criticism — it is a different problem, and it is the problem this paper addresses.

Persistence stability, image persistence, and well groups. Our certified-cycle count is a persistence quantity, and we say which one: writing $F_t = \{p \geq t\}$ for the closed superlevel filtration of the score map and $F_t^\circ = \{p > t\}$ for its open counterpart, the *persistence-form* count of §3.5 is

$$\operatorname{rank} H_1(F^\circ_{1/2+\varepsilon} \to F_{1/2-\varepsilon})$$

— by the standard rank formula, the number of points of p's superlevel persistence diagram with birth $> \frac{1}{2} + \varepsilon$ and death $< \frac{1}{2} - \varepsilon$, i.e. alive across the entire window $[\frac{1}{2} - \varepsilon, \frac{1}{2} + \varepsilon]$. The strict inequality at the source end is exactly CERT_FG $= \{p - \varepsilon > \frac{1}{2}\}$, and the target end is exactly ¬CERT_BG under the non-strict background test of §2; the headline enclosure count is this rank only when each cavity holds a single CERT_BG component (§3.5, §5.5). The stability theorem [8] recovers exactly this number as a lower bound: $\|p' - p\|_\infty \leq \varepsilon$ moves the diagram by at most ε in bottleneck distance, so every window-straddling point yields a hole in every reachable mask. Three things do not follow from stability. First, identity and localisation: a bottleneck matching is existential and non-canonical — it preserves the *count* of holes, not *which* holes; our certificate pins each feature by containment, with per-feature pixel support from the enclosure form, and its "same feature across masks" semantics is that of inclusion-induced maps — image persistence [9], the machinery also underlying Betti matching [26] — not a diagram metric. In particular the folklore reading "persistence exceeding 2ε implies the feature survives" is **false** at a fixed threshold: Figure 5's fillable ring has superlevel persistence $0.45 > 2\varepsilon = 0.2$, yet an $\varepsilon = 0.1$ perturbation destroys it — its bar, though long, does not straddle the window at $\frac{1}{2}$. Window-straddling, not persistence, is the correct criterion, and it is what we compute; §4 is the story of what happens when this distinction is missed. Second, completeness and witnesses: stability is a one-way inequality, whereas Proposition 1 characterises the reachable set exactly, so every non-certified hole ships with an explicit in-budget attack mask (filled at $M_{\max}$ or breached at $M_{\min}$). Third, scope: the connectivity class — designated endpoints staying in one component — plus coverage and abstention are not functionals of any persistence diagram (Appendix B gives two maps with identical superlevel diagrams and opposite connectivity verdicts), so no stability statement speaks to them. In the other direction stability is strictly broader where it applies: all thresholds at once, arbitrary filtrations, graceful degradation in Wasserstein distance; our construction is the single-threshold instantiation made exact — an ℓ_∞ ε-band *is* an ε-interleaving of superlevel filtrations. The closest prior notion is the **well group** of a level set [3, 11]: homology classes surviving every ε-perturbation — our universal quantifier. In this codimension-zero cubical setting the ε-well group of the $\frac{1}{2}$-superlevel set is exactly the image whose rank we compute; well groups are incomplete and uncomputable in general [13], while §3.5 is an operational, implementation-verified instance with localisation, witnesses, and non-homological companions.

Uncertain-field topology and three-valued abstraction. Mandatory critical points of uncertain scalar fields [17] compute, from per-vertex lower/upper bound fields, regions where critical points must occur in every realization — interval-bound "must" topology, shipped in TTK. They neither produce a thresholded-segmentation certificate nor characterise the reachable mask set exactly (Proposition 1), and provide no per-feature attack witnesses or soundness harness. More abstractly, the sandwich is a three-valued (Kleene) abstraction of the thresholded image, and the two extremes are the pessimistic/optimistic completions of three-valued model checking [5]; Lemma 1 is the image-topology instance of that classical duality — what this paper adds to it is the exactness of the abstraction (Proposition 1), the tightness of the two-mask check (§3.6), and the certified-class instantiation with its harness. A further adjacent line certifies downstream *classifiers* via Lipschitz control of persistence representations [1] — diagram-metric robustness of a prediction, not set-level feature survival of the segmentation; topology-*enforcing* segmentation [19] constrains the output's topology without certifying its robustness (enforcement is not certification); and conformal morphological margins [22] give distribution-free *statistical* coverage of the truth mask — complementary outer bounds to our worst-case machinery. To our knowledge, ours is the only method that is simultaneously worst-case, topological, about the segmentation output itself, and verified by checked predicates with attack witnesses rather than sampling (Appendix D).

Abstract interpretation. The sandwich is an interval abstract domain for the thresholding operator, in the sense of Cousot and Cousot [10]; Proposition 1 says the abstraction is exact for this (monotone) operator. The soundness harness is then best understood as *testing the abstract transformer against its concretisation* at the extremal points.

Benson's unconditional life. Benson [4] computes Go groups that are alive under any sequence of opponent moves, by a monotone fixpoint over "vital regions" — an adversarially quantified, purely combinatorial certificate. Our construction shares the shape: a monotone operator, an adversary quantified over an explicit reachable set, and a certificate that is checked, not sampled.

7 Limitations (the honest ledger)

1. **Coverage is the big number; connectivity is the honest one.** 94%/60% are coverage; the strict certified-connectivity fraction is always smaller and answers "does the grout certainly stay unbroken?" — on real p it is 0.766/0.343 at $\varepsilon = 0.10/0.40$ (a modest 1.2–1.7× gap), collapsing to an order of magnitude only on the near-tautological toy track (0.036 vs 0.500 at $\varepsilon = 0.40$), where the blurred maps fragment (§5.2).

2. **Score-map-level threat model.** Nothing here certifies the input→map pipeline (§1, §2); composition is an interface (§8), not a result.

3. **Small real study; non-vacuity is single-domain.** The *non-vacuity* headline rests on 30 crops, one task (mosaic grout), one frozen checkpoint; a larger real study would tighten that distribution. *Soundness* is validated more broadly — cross-domain on two retinal-vessel datasets (§5.7) — but we make no non-vacuity claim there (CHASE's cbrt maps are degenerate). The machinery itself is task-agnostic for binary maps thresholded at $\frac{1}{2}$.

4. **Cycle localisation granularity.** Persistence supplies the certified count; localisation uses the enclosure form (per-cycle generators are not extracted). The raw persistence count is unfiltered by ROI/scale floor and is reported as a cross-check column, not the headline.

5. **Toy track is nearly tautological** — caveat quoted verbatim in §5.3; we rely on it only for machinery validation, never for evidence about real-model behaviour.

Abstention is not the contribution. For a scalar binary score, the POSSIBLE band $|p - \frac{1}{2}| \le \varepsilon$ is a monotone transform of predictive entropy, so the abstain map by itself is equivalent to entropy thresholding and we claim no novelty for it. The contribution is what is *extracted* from the band — worst-case topological guarantees; entropy thresholding by itself certifies nothing about connectivity or holes. Figure 1 is the visual form of this distinction: the per-pixel margin calls nearly every pixel confident, while the per-feature certified budgets are graded and far sparser, because a feature's budget is the *minimum* over its wall and its interior — an aggregate no per-pixel map contains.

8 Future work

Composition with input-space certification. Any method producing per-pixel bounds $[l(x), u(x)]$ on the score map under input-space perturbations — randomized smoothing at segmentation scale [12], interval bound propagation [16] — supplies an interval field that replaces $[p - \varepsilon, p + \varepsilon]$ pointwise; Proposition 1 and every certificate in §3 apply unchanged to the resulting (per-pixel, non-uniform) sandwich. That composition would yield end-to-end certified topology. We state the interface and stop: it is not built, and nothing above claims it.

Graded certificates. The exact nesting of §5.4 means the family $\{\text{sandwich}(\varepsilon)\}$ is a filtration in ε; equivalently, (p, ε) defines an interval-valued image whose α-cuts are the sandwiches. A graded certificate — "this cycle survives to ε_1, that component to ε_2" — is then a persistence structure over the ε-axis itself. For cycles this is not hypothetical: each certified cycle already carries its exact maximal budget

$$\varepsilon^* = \min\left(p_{\text{birth}} - \tfrac{1}{2},\ \tfrac{1}{2} - p_{\text{death}}\right),$$

read off the same superlevel diagram that verifies the identity of §5.5. The construction connects to the graded fuzzy-partition representation lineage of the F-transform [23]. The per-feature budgets themselves are readable today and we visualise them (Figure 1); what is *not* built is the graded object proper — a persistence structure over the ε-axis with its own diagram, stability, and matching. That remains mathematical framing.

Multi-class segmentation. Per-class one-vs-rest score maps yield exact per-class sandwiches (Proposition 1 applies to each map separately); the joint labeling set embeds strictly into their product, so cross-class predicates — certified adjacency of two named regions, certified separation of class pairs — are sound but conservative when read off the product. The certificate types this enables (organ-touches-organ, phase contact) have no pixelwise analogue; we develop them separately.

Temporal. Per-frame certificates lift through the product lattice to certified topological *events* for free; certified *identity* of a feature across frames does not, and is the open problem.

Contingent topology. Intervals of the §3.5 filtration born or dying at the middle level flag holes present in *some but not all* reachable masks — a certified / contingent / absent triple per homology degree at zero extra compute; its per-mask quantifier semantics remain to be pinned down, so we report nothing under it yet.

9 Conclusion

An ℓ_∞ band around a frozen score map pins the reachable binary masks to a lattice interval *exactly* (Proposition 1), and over that interval a conjunction of monotone predicates is settled by its two extremes — both of them necessary, and not replaceable by any fixed budget of random masks (Lemma 1, §3.6). That is the whole engine; the certified components, tessera cores, unfillable holes, and unbroken segments are instantiations of it, each shipped with the adversary that would break it.

Two things we would ask a reader to carry away are corrections rather than claims. First, the instantiation is not mechanical: our own first cycle certificate was unsound, because H_1 is not monotone under foreground growth, and it survived an existing harness that simply did not know cycles were a class (§4). Second, coverage is not connectivity: they differ by 1.2–1.7× on real maps and by an order of magnitude on blurred ones, so we report them as two numbers and never one (§5.2). The certificate is also not a confidence map — a feature's budget is the minimum over its wall *and* its interior, an aggregate no per-pixel margin contains (Figure 1).

What we do not claim bounds what we do: nothing here certifies the input→score-map pipeline, and the non-vacuity numbers are one mosaic task and one frozen checkpoint. Soundness, though, held everywhere we attacked it — 16,675,936 checks on real mosaic crops and 10,578,388 across two retinal-vessel datasets, zero violations, including on maps degenerate enough that a certificate had every excuse to break.

10 Reproducibility and disclosure

Reproducibility. The experiment directory is registration-first: the registry records each hypothesis with its adversary, kill condition, and fixed conditions — the ε sweep, connectivity conventions, scale floor, and seeds — dated at the point of registration and before the corresponding numbers were produced. The registry ships with the code, so the registrations, and the order in which they were made, can be read directly. Corrections are appended as addenda that preserve the original claims verbatim rather than edited in place, and there are now two: the unsound β_1 cycle certificate of §4, and the refutation of the registered persistence/enclosure equivalence (§5.5). Both were found the same way — a registered claim checked against a case its own regression suite did not cover — and both are recorded with the construction that breaks them. The wording of every guarantee in Table 1 is locked in a guarantee-statement file that this paper's claims are checked against, and that file carries the same two revisions. Tables 2 and 3 are regenerated from the committed result files by `tables.py`; figures by `make_figs.py`; `check_numbers.py` traces every numeral in this paper to its source file or derivation. Test suite: 21/21, including the planted-unsound and counterexample regressions.

AI involvement. The experimental campaign was executed by an agentic AI system under human adjudication, with pre-registration (registry before code, claims locked before prose). The unsound cycle certificate of §4 was caught by an adversarial-verification pass using two independent AI reviewers; the finding, the halt, and the correction were adjudicated by the human author.

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A Per-crop distributions, and how the tables are rounded

Tables 2 and 3 report per-crop *means*; this appendix gives the underlying spread (Figure 11). The worst-case floors quoted in §5.1 are read off it, and they fall on *different* crops for the two quantities.

The tables themselves are complete for the swept grid; exact unrounded values are in the released results files (`results_real_hardened.json`, `results_toy_hardened.json`) and are regenerated by `tables.py`. Rounding convention, applied per column rather than per value: fractions to 3 decimals; per-crop mean counts to 1 decimal, except Table 3's two cycle columns, which carry 2 throughout because their values fall below 1 at the largest budgets.

B The counterexample constructions

Figure 5's grids are the regression-test arrays: a 40×40 field, background score 0.05, ring $8 < r < 14$ at score 0.95, interior $r \leq 8$ at score 0.5 (fillable case) or 0.05 (unfillable case), $\varepsilon = 0.1$. Under the sandwich: the ring is CERT_FG in both cases; the interior is POSSIBLE in the first (0.5 ± 0.1 straddles $\frac{1}{2}$) and CERT_BG in the second ($0.05 + 0.1 < \frac{1}{2}$). β_1(CERT_FG) $= 1$ in both; only the second survives the all-POSSIBLE →foreground extreme. Figure 8's pinched bar: a 30×60 field, bar rows 13–16, columns 5–54 (inclusive) at 0.95, with the two pinch columns 29–30 at 0.55 — above threshold (so the skeleton spans) but within $\varepsilon = 0.1$ of $\frac{1}{2}$ (so the two pinch pixels per row are POSSIBLE, and the spanning segment is not certified-connected).

Connectivity is not a diagram functional. Take three identical CERT_FG squares X, Y, Z over CERT_BG background and one CERT_FG bridge of fixed shape: joining X–Y (map A) or joining Y–Z (map B) produces identical superlevel persistence diagrams in every degree — the diagram is location-blind — yet with designated endpoints in X and Y, map A's pair is certified-connected while map B's pair is certainly separated in every reachable mask. No functional of the persistence diagram computes the connectivity class; that certificate is not a persistence statement.

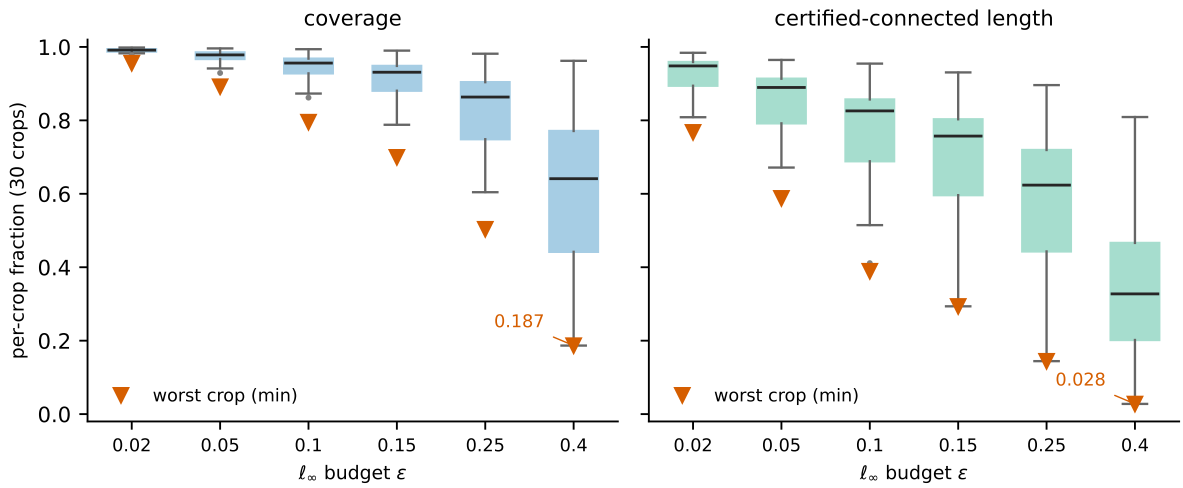

Figure 11: Per-crop distribution of coverage (left) and certified-connected length (right) across the 30 real crops at each ε (box = IQR, whiskers = range, dots = outliers). The worst crop (min, ▼) is marked: the cohort floors are coverage 0.187 and certified connectivity 0.028 at $\varepsilon = 0.40$ (each a minimum, on *different* crops), so the §5.1 headline is a floor over the cohort, not a mean pulled up by easy crops.

C Soundness-check accounting

Every emitted certified class is checked against 32 reachable masks: the two deterministic extremes plus 30 random reachable masks. The real-track total across 30 crops, the four certified classes (components, tessera cores, corrected cycles, and connectivity segments), and the six-point ε sweep is 16,675,936 individual checks — exactly $32 \times 521{,}123$ class instances — with 0 violations. The check for a foreground class asserts connectivity and presence in the minimal mask; for a background core, absence from the maximal mask; for a cycle, hole persistence at both extremes; for a connectivity segment, endpoint connectedness in the minimal mask.

Random-mask baseline (registered). To measure §3.6's tightness claim, `random_baseline.py` plants fillable square rings (interior area $= c$) and pinched bars (min-cut c verified exactly by max-flow) and scores the *semantic* violation events — hole destroyed, endpoints disconnected — under two samplers: the uniform reachable-mask model of §3.6 and the harness's own (per-mask $q \sim U(0.1, 0.9)$, iid per pixel). Detection probabilities are computed exactly (closed form for the ring; full pinch-block enumeration for the bar) and Monte-Carlo cross-checked; the committed `results_random_baseline.json` shows fair-coin ring detection equal to 2^{-c} exactly, and the harness sampler decaying exponentially too, but with a slower base — 0.9 rather than $\frac{1}{2}$, times a $1/(c+1)$ factor — an advantage of biased sampling that still vanishes as c grows, and every planted violation caught at exactly one extreme — rings at $M_{\max}$, bars at $M_{\min}$ — never the other (Figure 12).

D Related-work delta table

This is the paper's positioning claim in full: for each adjacent method, *what* it certifies, under *which* threat model, and whether the guarantee is worst-case, topological, and *checked* rather than sampled. The bottom row is INTACT, so the table is read as the delta between this paper and its neighbours. Every cell was verified against the primary source (staging notes committed in the repo); §6 gives the one-sentence version.

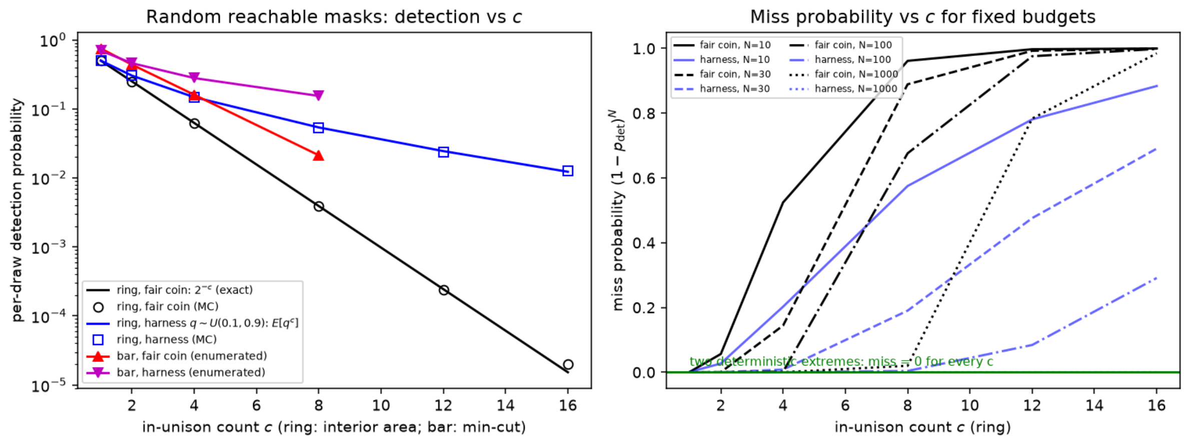

Figure 12: Random-mask baseline (registered). Left: per-draw detection of planted semantic violations vs the in-unison count c; exact laws, enumeration, and Monte-Carlo agree. Right: miss probability $(1 - p_{\mathrm{det}})^N$ at fixed budgets climbs toward one as c grows, while the two deterministic extremes catch every planted violation with certainty, each at exactly the predicted extreme.

Method	Output certified	Threat model	Worst-case?	Topological?	Checked vs sampled
SegCertify [12] / AdaptiveCertify [2]	per-pixel labels	input-space (smoothing)	probabilistic	no	sampled
Betti matching [26] + topology-aware losses	training loss on the learned map	none	no	yes (loss)	n/a (training)
Well groups [3, 11]	level-set homology rank of a fixed field	ℓ_∞ on the field	yes	yes	diagram read; uncomputable in general [13]
Mandatory critical points [17]	guaranteed critical-point regions	per-vertex interval bounds	yes	yes	tree pairing / visualization
Topology-guaranteed segmentation [19]	enforced connectivity/genus of the output	none	no (enforce $\neq$ certify)	yes	optimization
Conformal morphological sets [22]	statistical coverage of the truth mask	calibration exchangeability	probabilistic	no	calibrated margin
PH-representation robustness [1]	downstream label via persistence features	input Lipschitz	yes (label)	diagram-metric	Lipschitz bound
INTACT (this paper)	**four topological classes of the segmentation itself**	ℓ_∞ **on** p	**yes**	**yes**	**checked + adversarial harness with witnesses**

E Compute and runtime

All certification is CPU-only; a single desktop GPU (RTX 4080, 16 GB) is used once, for the frozen model's inference over the 30 crops. On one core, uncontended, the full per-unit cost (certification *plus* the soundness harness) is ≈51 s per mosaic (crop, ε) at the 32-mask setting and ≈104 s per retinal (image, ε) at 16 masks (≈36 s at 4). Certification proper is only 1–14 s of that; the soundness harness — thousands of certified classes × the mask budget — is the bulk, which is exactly why the two-extreme minimality of §3.6 matters in practice (the extremes are exhaustive, so the mask budget above two buys only cross-checks). The per-(unit, ε) cells are independent, so the whole study is embarrassingly parallel: the 30-crop mosaic run (`run_parallel.py`) and the retinal study (DRIVE via `drive_v2.py` and CHASE's 20-image × 3-normalization × 6-ε sweep via `chase_parallel.py`) each finish in minutes-to-a-couple-hours of wall time on the 24-thread machine (64 GB RAM). A near-linear implementation of the connectivity segment graph (§3.4) was load-bearing: it cut dense-vessel certification from >20 minutes to ≈14 s per (image, ε), without which the retinal study was intractable.